A069 v0.6.2 - Dual-Trefoil High-Resolution GP Campaign

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Research scope

The user-requested campaign includes 160 cubed grids for both historical trefoil seeds: Fourier A (2015 p.32) and Torus B (2015 p.66). Optional grid refinements are B192, B256, A320 and A512. The goal is to test knot-line reconstruction and dynamical topology, not to fit a Swirl Clock or claim an already established internal eigenmode.

Critical preregistered core-readiness inequality

At box length L=10, G32 requires xi*N/L >= 4, xi*kappa_max <= 0.30 and xi/d_nonlocal <= 0.30. The nonlocal gap is a screened segment-distance estimate; it is not a mathematically certified tube reach. Thus even G32 PASS is only a preflight screen.

Fourier A at N=160 requires xi >= 0.25 to resolve four cells per core, but xi <= 0.08524 from the nonlocal separation estimate. No value of xi satisfies both constraints. The requested A160 run is therefore diagnostic-only. For A, the first even grid meeting this screen is N=470; A512 with xi=0.08 is the selected high-memory candidate. Torus B at N=160 with xi=0.28 passes the screening inequalities.

Case	N x N x N	xi	cells/xi	xi*kappa	xi/gap	G32	Peak RAM*
A160	160	0.08	1.28	0.134	0.282	FAIL	4.0 GiB
B160	160	0.28	4.48	0.286	0.246	SCREEN PASS	4.0 GiB
B192	192	0.28	5.38	0.286	0.246	SCREEN PASS	5.8 GiB
B256	256	0.28	7.17	0.286	0.246	SCREEN PASS	11.7 GiB
A320	320	0.08	2.56	0.134	0.282	FAIL	21.3 GiB
A512	512	0.08	4.10	0.134	0.282	SCREEN PASS	82.8 GiB

* RAM figures are conservative forecasts rather than measured resident peaks; free physical memory and temporary FFT arrays are checked prior to execution. N512 may require about 96-128 GiB of installed RAM, large storage and many CPU-hours.

Actual computations performed in this build environment

46/46 inherited and new Python regression tests have passed. Two short actual 3D Gross-Pitaevskii smoke evolutions were performed at N=32 (Fourier A) and N=48 (Torus B). They exercised initialization, split-step time propagation, conservation diagnostics, subgrid face-zero tracking, multi-angle knot projection and immutable JSON output; a strict continuation/checkpoint test also completed. Both smoke runs intentionally fail G32 due to insufficient core resolution.

The B48 smoke run recovered a closed final polygon with numerical Fox determinant three, but its initial graph was ambiguous. This is a useful topological projection diagnostic only, NOT a proof of ambient isotopy during evolution. No 160-cubed or larger 3D GP dynamics has been executed in this environment: the conservative RAM guard fails here.

GP coupling and physical core-scale compatibility

A scientific defect in the earlier GP preflight has been isolated: the v0.5.0 normalized equation fixes the GP nonlinearity to mu=1, whereas xi was varied only in the initial amplitude profile. Those profiles are not generally stationary and can broaden to the bulk healing scale. In v0.6.2 we evolve i dPsi/dT = -0.5 Laplacian(Psi) + mu (|Psi|^2 - 1) Psi, setting mu=1/(2 xi^2) to match the exact planar black-soliton width

convention used by the initializer tanh[d/(sqrt(2)xi)]. This makes initialization and GP bulk coupling dimensionally consistent. It does NOT establish a stationary curved 3D vortex core. The flag --legacy-normalized-gp retains mu=1 strictly for historical regression.

The B160/B192/B256 ladder at fixed xi=0.28 also holds mu=6.377551 constant, allowing spatial refinement of one GP model. Comparing xi=0.28 against 0.24 changes mu and is an independent physical-model/core sensitivity test, not pure continuum grid convergence.

Scientific gate state

G32: B160/B192/B256 and A512 pass the preregistered geometry screening at the stated xi; A160/A320 fail. This is neither certified reach nor vortex stability.

G33: subgrid localization, dual graph diagnostics and five numerical projection directions are implemented. Single-snapshot projection consistency does not certify a time-continuous trefoil.

G34: UNRESOLVED - no complete cross-grid/time/core/domain convergence, ambient-isotopy continuity or independently verified reconnection event.

G35: UNRESOLVED - no stationary or relative-periodic knot, symmetry-reduced 3D Floquet/BdG eigenmode or blinded frequency holdout.

Method and computational integrity

The GP equation is i dPsi/dT = -1/2 Laplacian(Psi) + (|Psi|^2 - 1) Psi, in dimensionless units. The public A069 v0.5.0 source and initializer remain SHA-256 pinned (bb8e1d392e3e72c0c2caee22df406469f9f3de5504296da45cedc6fda8d04b6e). The v0.6.1 lattice/subgrid modules are inherited with a versioned configurable-coupling 3D GP stepper and new non-mutating development overlays. The code computes evolving order-parameter fields in all three spatial coordinates, not full incompressible SST Euler dynamics. It excludes gold-ratio constraints, measured clock values, Compton-frequency fits, Lorentz input and legacy visual KnotPlot relaxations.

Windows campaign execution

Install requirements, unpack pinned solver, run 38 regression tests and print the grid/memory plan:

```
py -m pip install -r requirements.txt
py bootstrap_upstream.py
py -m pytest -q tests
py plan_v062.py
run_preflight_160.cmd
run_optional_high_res.cmd
```

The supplied scripts do not run A512 automatically. Source ZIP and output JSON are separate versions; neither FREEZE, BASIC, C++/SYCL parity nor official scientific BLIND reveal was performed here.

Bottom line

A069 v0.6.2 is a tested, runnable high-resolution campaign package, NOT a new scientific result from a 160-cubed run. The next valid decision depends on new user-side convergence data, particularly B160/B192/B256 first. The intrinsic Swirl Clock eigenfrequency remains UNRESOLVED.